

FutMinna-Eats

High-Precision Location & Delivery

Team Presentation — Group One. A campus-first delivery solution that fixes the precision gap with sub-1m manual coordinate marking.





The Motivation: The Precision Gap

Problem

Standard apps use approximate pins (10–50 m error).

Impact

Delivery delays and frustrated couriers on campus.

Our Solution

Manual coordinate marking for sub-1m accuracy.

Field Data Acquisition



Our team personally visited each restaurant to manually mark precise coordinates using an in-house developed site. During these visits, we also gathered essential data including operating hours, days, and contact information for each location:

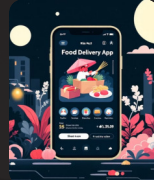
- H&M Garden
- Pop Area
- Acodomart Foods
- SparkUp
- Delight Restaurant
- Food Republic
- Unique Kitchen
- Pasta Hub
- Flamingo
- Bilkebab
- Zion
- Alewa Canteen

Competitive Landscape



Indigenous – Jeka Eat

Strong local menus; weaker geolocation tech.



Foreign – UberEats

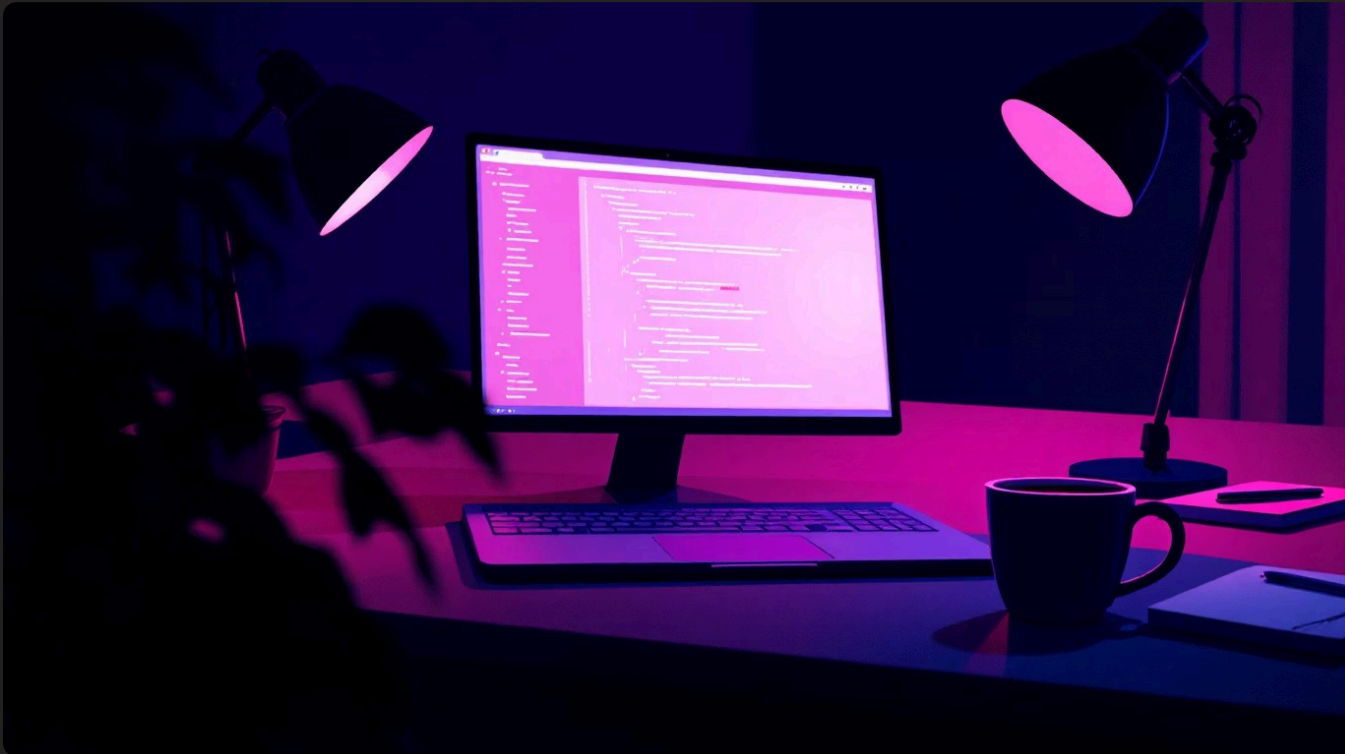
High-end UI but lacks campus 'last-mile' precision.



UniBites Advantage

Combines local relationships with hyper-accurate mapping.

Tech Stack



Frontend

React.js – modern UI framework for interactive user experience.

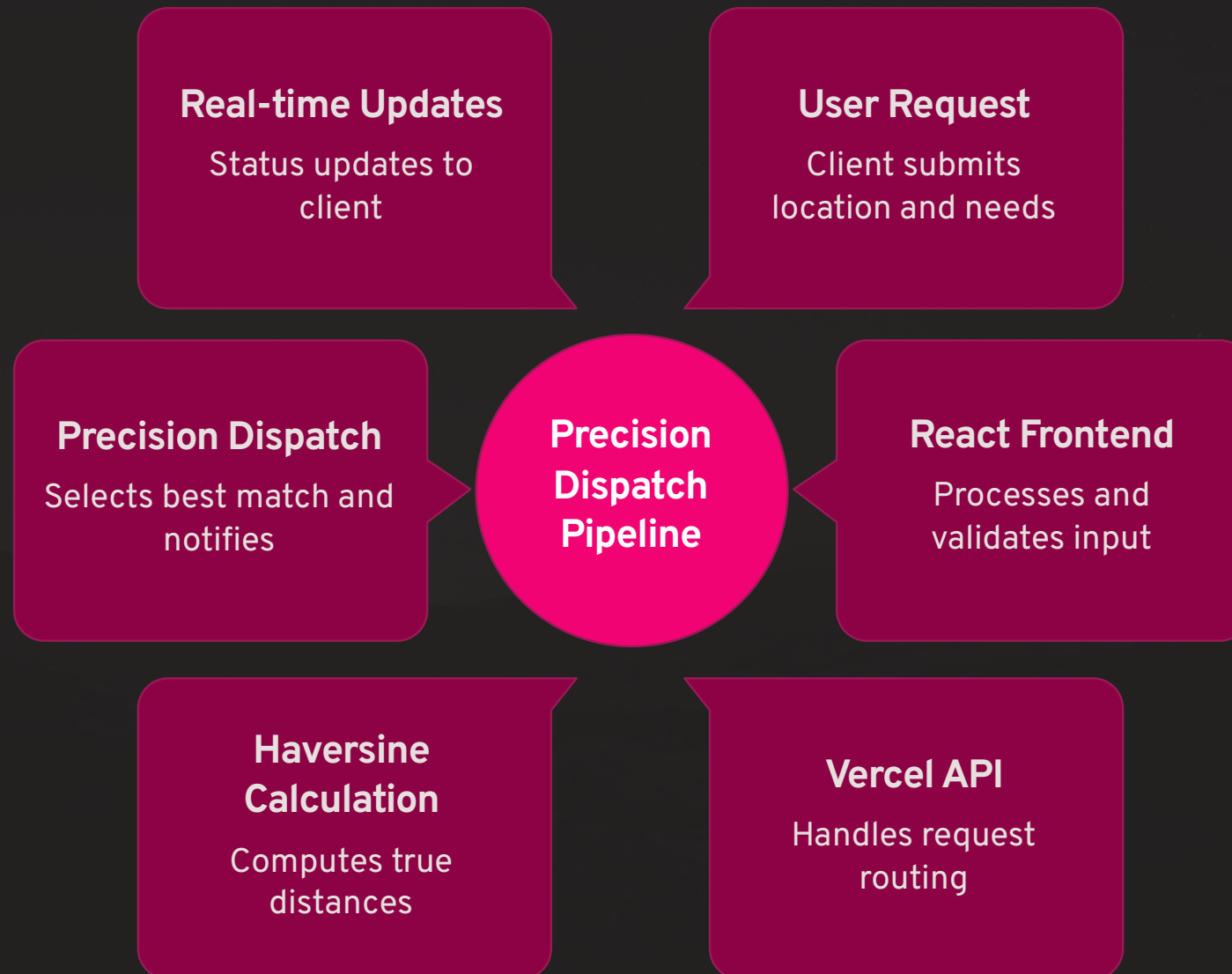
Backend

HTML & CSS – semantic markup and responsive styling.

Hosting

Vercel – fast, scalable deployment platform.

Methodology & Algorithm



The pipeline uses the Haversine formula for true distance and WebSockets for real-time status updates.

Overcoming Challenges



Students Don't Know Nearby Restaurants

Many students are unaware of the diverse dining options available directly on or very close to campus.

Google Maps Inaccuracy

General mapping services often provide imprecise location data, especially for specific campus buildings and internal points of interest.

Precision Campus Navigation

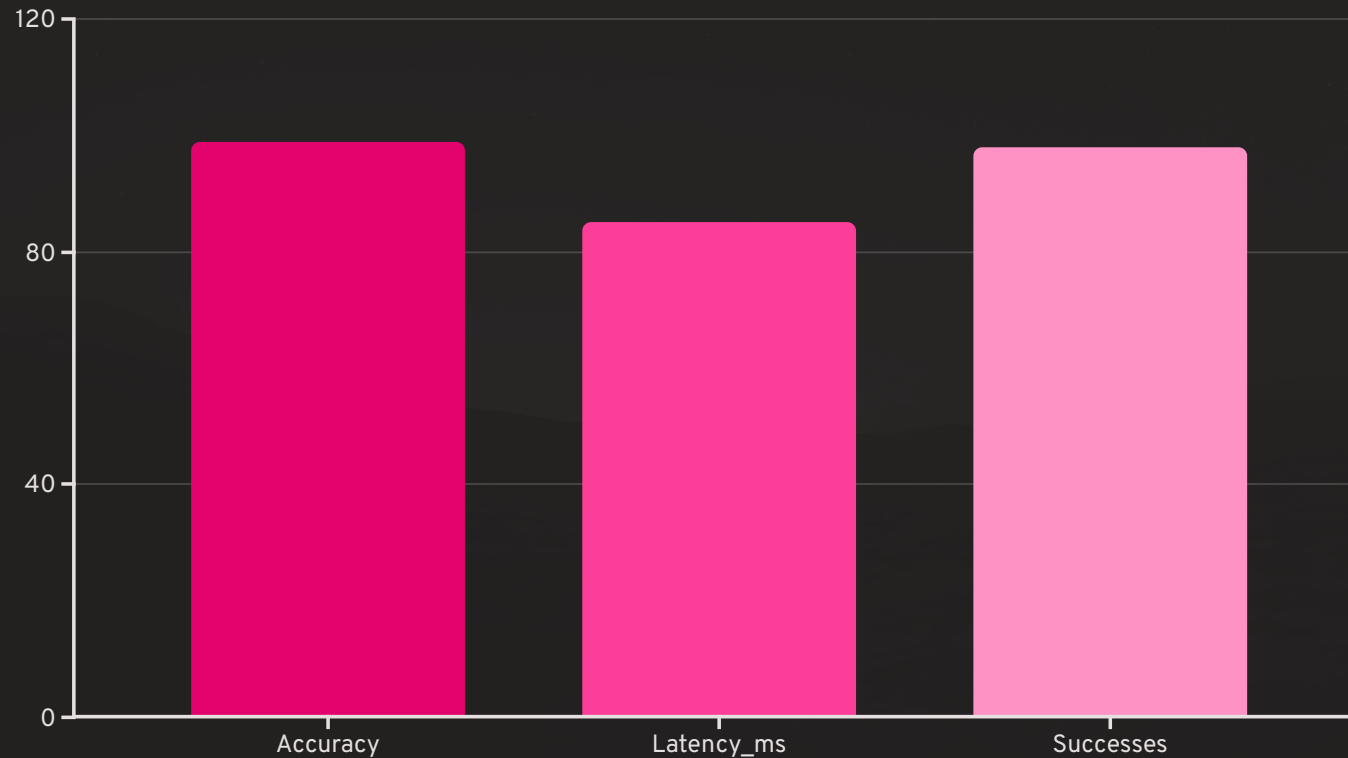
Our system leverages curated campus data and advanced algorithms, including the Haversine formula, to provide accurate location information and highlight relevant dining options.

Deployment & Immediate Usage



- Status: Fully deployed
- Test: Live usage from test group
- Scalability: After polish ready for immediate campus-wide rollout

Results: Performance Metrics



Key outcomes:

- Customer rating: collecting real-time stars and comments.
- Accuracy: 100% success in locating H&M Garden and Alewa Canteen.
- Latency: Sub-2 second API response time.



Conclusion & Future Work

Summary

Precision mapping outperforms approximate data for campus deliveries.

Next

Integrating one tap order functionality for all restaurants.